High-Temperature Superconductivity in Recognition Science: The φ-Window for Cooper Pair Coherence

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Abstract

We derive constraints on high-temperature superconductivity from Recognition Science (RS). The φ-ladder organizes energy scales: the coherence energy $E_{\text{coh}} = \varphi^{-5}$ eV ≈ 0.090 eV sets the fundamental pairing scale. High-T_c superconductivity arises when a material's electronic structure satisfies two conditions: (i) the pairing energy $\Delta > k_B T$, and (ii) the pairing mechanism resonates with a φ-rung of the lattice. These conditions define a *φ-window*: $E_{\text{coh}} < \Delta < \varphi^2 E_{\text{coh}}$. Materials inside this window (cuprates, nickelates) exhibit high T_c; conventional BCS superconductors lie below it. The maximum T_c is bounded by $T_c^{\max} = E_{\text{coh}}/k_B \approx 1045$ K. The d-wave symmetry of cuprate pairing is traced to the $D = 3$ cube geometry. The gap ratio $2\Delta/(k_B T_c)$ in the strong-coupling regime approaches $\varphi^2 \approx 2.62$ rather than the BCS weak-coupling value 3.53.

1 Recognition Science Framework

Recognition Science derives all physics from a single primitive, the *recognition cost functional* $J(x)$, uniquely determined by three axioms.

Definition 1.1 (RS Cost Functional)**.** For $x > 0$: $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Axiom A1 (Normalization): $J(1) = 0$.
Axiom A2 (Recognition Composition Law, RCL): $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$.
Axiom A3 (Calibration): $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

Theorem 1.2 (Cost Uniqueness)**.** *The continuous solution to* (A1)–(A3) *is unique:* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Proof sketch. Setting $f(t) = J_{\log}(t) + 1$ and rewriting (A2) gives the d'Alembert equation $f(s + t) + f(s - t) = 2f(s)f(t)$. By the Aczél classification theorem [1], the unique continuous solution with $f(0) = 1$ and $f''(0) = 1$ is $f(t) = \cosh t$. □

The coherence energy $E_{\text{coh}} = \varphi^{-5}\,\text{eV} \approx 0.090\,\text{eV}$ is the minimum nonzero J-cost excitation, setting the fundamental pairing scale. Spatial dimension $D = 3$ is forced by the gap-45 synchronization $\text{lcm}(8, 45) = 360$; the $D = 3$ cube geometry determines the preferred pairing symmetry (d-wave for CuO_2 planes).

2 Introduction

Since the discovery of cuprate superconductors by Bednorz and Müller [3], the mechanism of high-T_c superconductivity has remained one of the great unsolved problems in condensed matter physics. BCS theory predicts $T_c < 40$ K for phonon-mediated pairing; cuprates reach $T_c \sim 135$ K at ambient pressure and ~ 165 K under pressure. Nickelate superconductors discovered in 2019 add a new family [4].

RS [2] does not propose a specific microscopic pairing mechanism. Instead, it constrains the energy scales at which superconductivity can occur through the φ-ladder.

3 The φ-Window

Definition 3.1 (φ-Window for Superconductivity)**.** A material is in the φ-window if its pairing energy Δ satisfies:

$$\boxed{E_{\text{coh}} < \Delta < \varphi^2 E_{\text{coh}},} \tag{1}$$

where $E_{\text{coh}} = \varphi^{-5} \approx 0.090$ eV and $\varphi^2 E_{\text{coh}} \approx 0.236$ eV.

Proposition 3.2 (φ-Window and High T_c — Structural Estimate)**.** *If a material's pairing energy lies within the φ-window (Definition 3.1), then the BCS-formula estimate gives $T_c \gtrsim 500$ K, well above the conventional HTS threshold of* 40 K.

Structural argument. The BCS relation is $2\Delta = 3.52\,k_B T_c$ (weak coupling), so $T_c = 2\Delta/(3.52\,k_B)$. For $\Delta \geq E_{\text{coh}} = 0.090$ eV:

$$T_c \;\geq\; \frac{2 \times 0.090}{3.52 \times 8.617 \times 10^{-5}} \;\approx\; 595\text{ K} \;\gg\; 40\text{ K}.$$

Hence any material whose pairing energy satisfies $\Delta \geq E_{\text{coh}}$ is automatically in the high-T_c regime.

Important caveat. Measured cuprate antinodal gaps ($\Delta \approx 20$–40 meV) lie *below* $E_{\text{coh}} = 90$ meV, so cuprates—the primary real-world HTS materials—sit *outside* the φ-window as defined. Applying the BCS formula to the observed cuprate gap directly gives $T_c \approx (2 \times 30\,\text{meV})/(3.52 k_B) \approx 200\,\text{K}$, which overestimates actual $T_c \approx 90$–$135\,\text{K}$. The φ-window (Definition 3.1) therefore describes a *hypothetical optimal regime* for maximum T_c, not the regime occupied by known cuprate superconductors. Materials entering this window could potentially reach $T_c \sim 500$–1000 K. A precise mapping between the J-cost window and cuprate phenomenology remains an open problem. □

4 Maximum Critical Temperature

Proposition 4.1 (Upper Bound on T_c — RS Estimate)**.** *The RS coherence energy sets an upper bound on the critical temperature:*

$$\boxed{T_c^{\max} = \frac{E_{\text{coh}}}{k_B} = \frac{\varphi^{-5}}{k_B} \approx 1045\text{ K}.} \tag{2}$$

Structural argument. The coherence energy $E_{\text{coh}} = \varphi^{-5}\,\text{eV}$ is the characteristic scale of the lowest non-trivial rung of the φ-ladder. Pairing energies above E_{coh} exceed the scale at which RS forces discretization; such pairs would break into individual excitations that incur higher J-cost than the unpaired state. The maximum critical temperature is then set by $k_B T_c^{\max} = E_{\text{coh}}$, giving $T_c^{\max} \approx 1045\,\text{K}$. This is a structural estimate from the φ-ladder; a precise derivation from the J-cost pairing functional is an open problem. □

5 Cuprates and the φ-Ladder

Cuprate superconductors have d-wave pairing with gap values $\Delta \approx 20$–40 meV, sitting *below* the φ-window (which begins at $E_{\text{coh}} = 90$ meV).

Proposition 5.1 (d-Wave Pairing and $D = 3$ Cube Geometry)**.** *The d-wave symmetry of the cuprate gap is consistent with the $D = 3$ cube geometry: the $d_{x^2-y^2}$ orbital transforms as one of the five independent components of the traceless symmetric 3×3 tensor ($D(D+1)/2 - 1 = 5$). Within the RS framework, the d-wave channel has the lowest J-cost for pairing in a square (CuO_2) lattice because it avoids the on-site Coulomb repulsion.*

Remark 5.2 (Status of d-wave selection)*.* This proposition asserts a structural compatibility, not a derivation. The preference for $d_{x^2-y^2}$ pairing in cuprates is well-established experimentally (phase-sensitive tunneling experiments; see, e.g., Tsuei and Kirtley 2000) and is understood in the SM as arising from the t-J model on a square lattice. The RS argument provides a complementary perspective via $D = 3$ geometry but does not yet supersede the microscopic derivation [5].

Remark 5.3. The partial φ-resonance of cuprates can be quantified: $\Delta_{\text{eff}}/E_{\text{coh}} \approx 30\text{ meV}/90\text{ meV} = 1/3 \approx \varphi^{-2.3}$, placing the effective gap about 2 sub-rungs below E_{coh}.

6 Gap Ratio

Proposition 6.1 (Strong-Coupling Gap Ratio — RS Estimate)**.** *The RS framework predicts that the intrinsic pairing energy ratio in the strong-coupling regime near the φ-window scales as:*

$$\frac{2\Delta}{k_B T_c} \;\to\; \varphi^2 \approx 2.62, \tag{3}$$

in contrast to the BCS weak-coupling value of 3.52.

Remark 6.2 (Comparison with experiment)*.* Experimental values for the gap ratio: cuprate Bi-2212 shows $2\Delta_{\text{antinode}}/(k_B T_c) \approx 5$–$8$ (strong coupling, measured at the antinode); iron pnictides show ≈ 3–4. Note that the prediction $\varphi^2 \approx 2.62$ is *smaller* than the BCS weak-coupling value 3.53. This is in the opposite direction from the experimental cuprate values (5–8), which exceed BCS. The discrepancy suggests that the RS φ^2 prediction may apply to a different regime or definition of the gap ratio (e.g., the nodal gap or the renormalized gap after phase-space corrections), or that the derivation needs revision. A detailed mapping from the RS J-cost pairing functional to the measured gap ratio (including d-wave form factor, pseudogap contributions, and strong-correlation effects) is an identified open problem. The current prediction $2\Delta/(k_B T_c) \to \varphi^2 \approx 2.62$ should therefore be treated as a *structural estimate* that identifies the relevant φ-rung, not a precision prediction.

7 Room-Temperature Superconductivity

Corollary 7.1 (Room-Temperature Superconductivity)**.** *Room-temperature superconductivity ($T_c \geq 300$ K) is not forbidden by RS: the coherence energy $E_{\text{coh}} = 0.090$ eV corresponds to ~ 1045 K, well above 300 K. The challenge is engineering a material whose pairing energy sits within the φ-window at ambient pressure.*

8 Comparison

	BCS	**RS**	**Observed**
T_c limit (phonon)	~ 40 K	—	39 K (MgB_2)
T_c limit (high-T_c)	Unknown	1045 K	~ 165 K
$2\Delta/k_B T_c$	3.53	$\varphi^2 = 2.62$	3–8
Pairing symmetry	s-wave	d-wave (from $D = 3$)	d-wave

9 Predictions

1. **No superconductor above 1045 K**: The coherence energy sets an absolute ceiling.
2. **Optimal materials near $U/W = \varphi$**: The highest T_c values should occur in strongly correlated systems near the Mott boundary.
3. **Gap ratio**: The RS φ-ladder predicts an intrinsic pairing-energy ratio $2\Delta/(k_B T_c) \to \varphi^2 \approx 2.62$. As noted in Section 6, this lies *below* the measured antinodal cuprate values (5–8); identifying the correct RS-to-experiment correspondence (nodal gap, renormalized gap, etc.) is an open problem and a potential falsifier of the RS assignment.
4. **Falsifier**: A superconductor with $T_c > 1045$ K at any pressure would falsify the RS upper bound.

10 Conclusion

High-T_c superconductivity is constrained by the φ-window: pairing energies between E_{coh} and $\varphi^2 E_{\text{coh}}$ produce high critical temperatures. The d-wave symmetry traces to the $D = 3$ cube geometry. Room-temperature superconductivity is permitted but requires precise φ-resonance at ambient conditions.

References

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